

TripWiz — Folder 11: System Administrator Manual

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How to read this manual

This is a **completely separate manual** from the End-User Manual (Folder 07) and the Admin Portal Manual (Folder 08). Its sole audience is the **System Administrator** — the technical operator responsible for deploying, configuring, monitoring, and maintaining the TripWiz **AWS serverless infrastructure** itself.

***This is not the in-app Admin Portal.** Folder 08 already documents the in-app Admin Portal (a TripWiz application feature used to moderate users, trips, and trending content through the TripWiz UI). This manual is for the person who keeps the underlying AWS stack running — deploying code, configuring services, rotating secrets, monitoring logs, and responding to infrastructure incidents. The two roles may be the same person on a small team, but the responsibilities and tools are entirely different.*

Every section below is a numbered, step-by-step walkthrough referencing the actual infrastructure declared in `backend/infra/template.yaml` (the AWS SAM template that defines the entire TripWiz backend). Wherever the AWS Management Console is the relevant tool, a placeholder of the form `[INSERT SCREENSHOT: ...]` marks exactly where to capture and paste a console screenshot before this manual is finalized.

***Note on supplementary materials:** As with the End-User Manual, any future video walkthroughs or in-app help content for administrators are **supplementary only**. Per the course guidelines, they do not replace this written manual, which remains the authoritative operations reference for the system.*

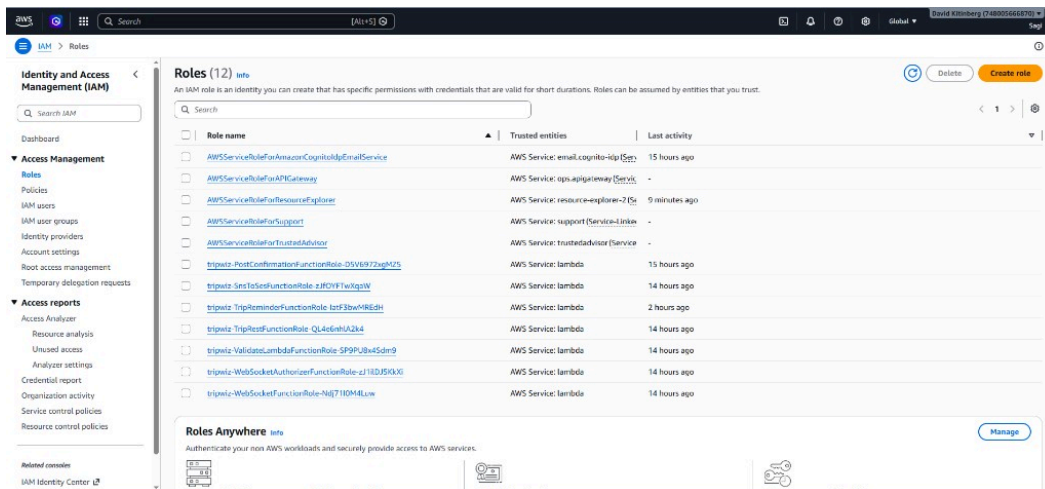
1. Introduction & Prerequisites

1.1 Who Should Use This Manual

This manual is intended for a **technically proficient operator** — a DevOps engineer, backend developer, or IT administrator — responsible for the AWS account hosting TripWiz. Familiarity with AWS fundamentals (IAM, CloudFormation, Lambda) and basic command-line usage is assumed.

1.2 Required AWS Account Access

1. You will need an **IAM user or role** with sufficient permissions to manage: CloudFormation/SAM stacks, Lambda, API Gateway, DynamoDB, Cognito, S3, Bedrock, Amazon Location Service, SES, SNS, EventBridge, Secrets Manager, and CloudWatch.
2. For initial setup and ongoing maintenance, **AdministratorAccess** (or an equivalently-scoped custom policy covering the services above) is recommended for the operator's IAM identity.



- Confirm you can sign in to the **AWS Management Console** for the account hosting TripWiz, and note the **AWS region** the stack is deployed to (TripWiz is configured for **US East (N. Virginia)** — **us-east-1** — by default).

1.3 Required Local Tools

Install the following on your operator workstation before attempting any deployment or maintenance task:

- AWS CLI v2** — for authenticating to AWS and running ad-hoc commands.
- AWS SAM CLI** — for building and deploying the serverless stack (`backend/infra/template.yaml` uses the `AWS::Serverless-2016-10-31` transform).
- Node.js 24.x** — matches the Lambda runtime (`nodejs24.x`) declared in the template's `Globals` section, ensuring local builds match the deployed environment.
- Git** — to pull the latest TripWiz source code from the repository.

Run `aws configure` to set up your CLI credentials and default region before proceeding.

1.4 Repository Layout — Where Things Live

Path	Purpose
<code>backend/infra/template.yaml</code>	The SAM template — the single source of truth for all AWS infrastructure
<code>backend/src/</code>	Lambda function source code (one subfolder per function)
<code>frontend/amplify.yaml</code>	Build specification for the AWS Amplify-hosted frontend
<code>06.md</code>	The full AWS cost estimate for this architecture (cross-referenced in Section 10)

2. Deploying & Updating the System

2.1 Building the Stack

- Open a terminal and navigate to the `backend/infra/` directory.
- Run:

```
sam build
```

This packages each Lambda function (from `CodeUri: ../src/`) and prepares the CloudFormation template for deployment.

3. Confirm the build completes without errors — any syntax issues in the Lambda source code will surface here before you attempt a deployment.

2.2 Deploying for the First Time

1. Run:

```
sam deploy --guided
```

2. The guided deployment will prompt you for:

- **Stack name** (e.g., `tripwiz-prod`)
- **AWS Region** (e.g., `us-east-1`)
- **Parameter values** — the template defines four parameters you must supply:

Parameter	Description	Example
<code>AdminEmail</code>	Address that receives TripWiz alert emails (must be SES-verified)	<code>admin@yourdomain.com</code>
<code>SourceEmail</code>	SES-verified sender address for outgoing TripWiz emails	<code>noreply@yourdomain.com</code>
<code>AdminBootstrapEmails</code>	Comma-separated emails granted admin access before Cognito group sync	<code>admin@yourdomain.com</code>
<code>ReminderDays</code>	Days before trip start to send reminder emails (default: <code>2</code>)	<code>2</code>

```
cd backend/infra
sam build
sam deploy
```

Deploying with following values

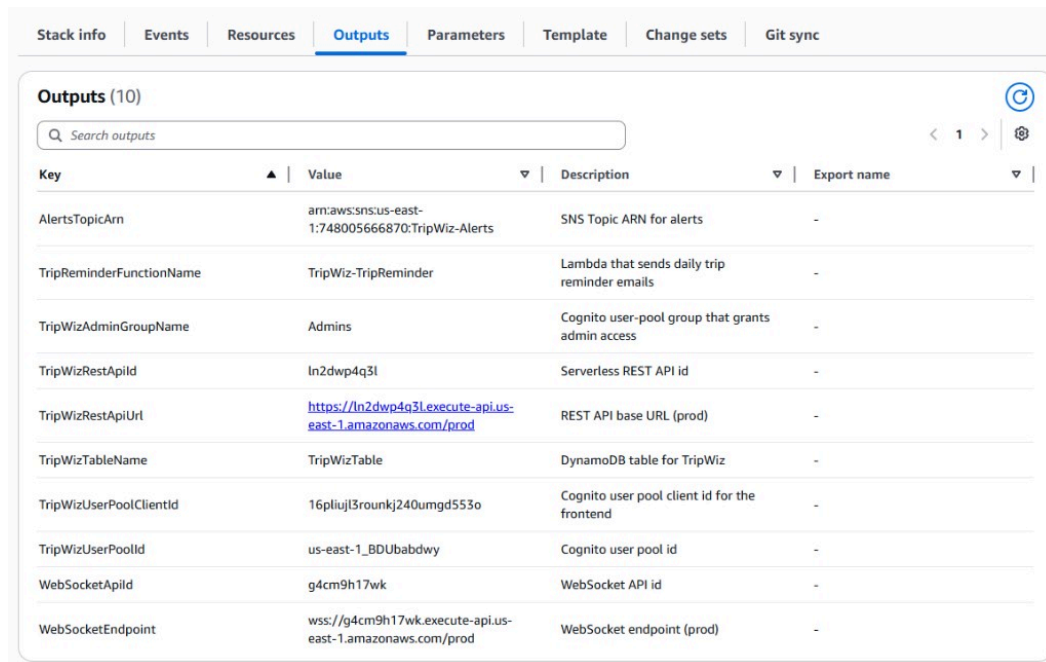
=====

```
Stack name           : tripwiz
Region               : us-east-1
Parameter overrides  : {
  "AdminEmail": "amituniversitymail@gmail.com",
  "SourceEmail": "amituniversitymail@gmail.com",
  "AdminBootstrapEmails": "amituniversitymail@gmail.com,davidkitinberg@gmail.com",
  "FrontendUrl": "https://main.d1qe3fil02rb8e.amplifyapp.com"
}
```

No changes to deploy. Stack tripwiz is up to date

3. Confirm changes when prompted, and allow the deployment to proceed. SAM will create a CloudFormation **change set** and apply it.

- Once complete, open the **AWS CloudFormation console** and locate your stack to review its **Outputs** tab — this lists critical values you'll need for frontend configuration and ongoing operations (REST API URL, WebSocket endpoint, Cognito User Pool ID and Client ID, DynamoDB table name, SNS topic ARN).



Stack info Events Resources Outputs Parameters Template Change sets Git sync				
Outputs (10)				
Q Search outputs				
Key	Value	Description	Export name	
AlertsTopicArn	arn:aws:sns:us-east-1:748005666870:TripWiz-Alerts	SNS Topic ARN for alerts	-	
TripReminderFunctionName	TripWiz-TripReminder	Lambda that sends daily trip reminder emails	-	
TripWizAdminGroupName	Admins	Cognito user-pool group that grants admin access	-	
TripWizRestApiId	ln2dwp4q3l	Serverless REST API id	-	
TripWizRestApiUrl	https://ln2dwp4q3l.execute-api.us-east-1.amazonaws.com/prod	REST API base URL (prod)	-	
TripWizTableName	TripWizTable	DynamoDB table for TripWiz	-	
TripWizUserPoolClientId	16plujl3rounkj240umgd553o	Cognito user pool client id for the frontend	-	
TripWizUserPoolId	us-east-1_BDUbabdwy	Cognito user pool id	-	
WebSocketApiId	g4cm9h17wk	WebSocket API id	-	
WebSocketEndpoint	wss://g4cm9h17wk.execute-api.us-east-1.amazonaws.com/prod	WebSocket endpoint (prod)	-	

2.3 Deploying Updates

- After pulling the latest code changes, repeat the build step:

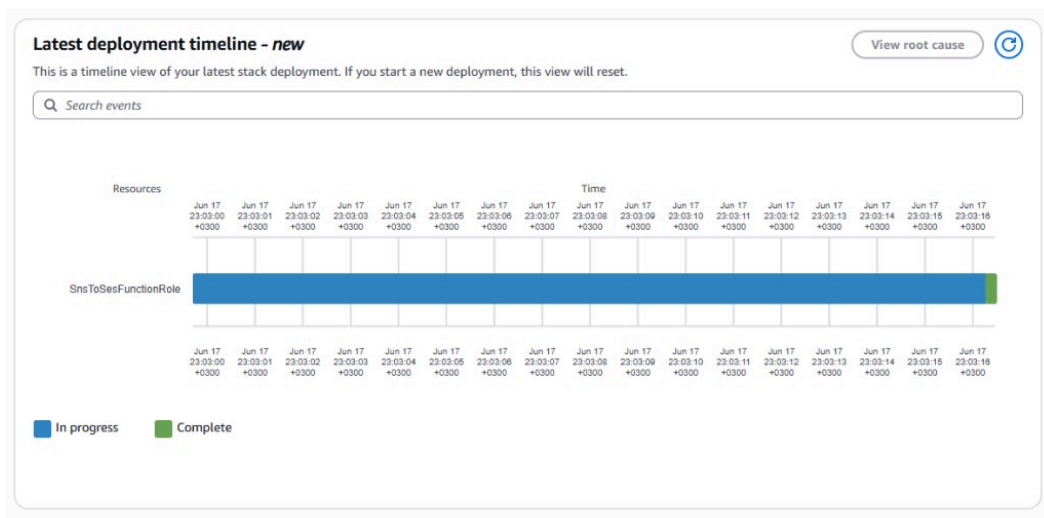
```
sam build
```

- Deploy the update:

```
sam deploy
```

(Once a `samconfig.toml` exists from your first guided deployment, subsequent deploys reuse the saved configuration without re-prompting.)

- Monitor the **CloudFormation console** to watch the stack transition through `UPDATE_IN_PROGRESS` to `UPDATE_COMPLETE`. If a deployment fails, CloudFormation automatically rolls back to the last known-good state — review the **Events** tab to diagnose the failure reason.



2.4 Required Post-Deployment Steps

The template provisions two **empty** Secrets Manager secrets that must be populated manually after the first deploy (see Section 8 for details):

1. TripWiz/OpenWeatherApiKey — your OpenWeatherMap API key
2. TripWiz/MappingApiKey — your mapping service API key (if using a third-party provider alongside Amazon Location Service)

Important: TripWiz will not function correctly until these secrets are populated. Do not skip Section 8 after your first deployment.

3. Managing Compute & APIs

3.1 The Seven Lambda Functions

TripWiz's entire backend runs on seven Lambda functions, all using the `nodejs24.x` runtime. Open the **AWS Lambda console** and locate each by name:

Function Name	Source Folder	Role
TripWiz-Trips	rest_handlers/trips.js	Main REST API handler — trips, places, admin routes, user prefs
TripWiz-Validate	validate_lambda/index.js	On-demand + nightly weather validation
TripWiz-WebSocketHandler	ws_handler/index.js	Real-time co-editing WebSocket router
TripWiz-WebSocketAuthorizer	ws_authorizer/index.js	JWT validation for WebSocket \$connect
TripWiz-TripReminder	trip_reminder/index.js	Daily scheduled trip-reminder emails
TripWiz-SnsToSes	sns_to_ses/index.js	Converts SNS weather alerts into admin emails

TripWiz-PostConfirmation	post_confirmation/index.js	Cognito post-confirmation trigger — seeds default notification prefs (notifyTrips: true) for every newly confirmed user
--------------------------	----------------------------	---

Functions (7)
Last fetched 6/18/2026, 1:45:56 PM
Actions
Create function

Functions table allows you to query function details. Use Ctrl + K to start a in-console resource search.

☐	Function name	Description	Package type	Runtime	Type	Last modified
☐	TripWiz-TripReminder	-	Zip	Node.js 24.x	Standard	15 hours ago
☐	TripWiz-SnsToSes	-	Zip	Node.js 24.x	Standard	15 hours ago
☐	TripWiz-WebSocketAuthorizer	-	Zip	Node.js 24.x	Standard	15 hours ago
☐	TripWiz-PostConfirmation	-	Zip	Node.js 24.x	Standard	15 hours ago
☐	TripWiz-WebSocketHandler	-	Zip	Node.js 24.x	Standard	15 hours ago
☐	TripWiz-Trips	-	Zip	Node.js 24.x	Standard	15 hours ago
☐	TripWiz-Validate	-	Zip	Node.js 24.x	Standard	15 hours ago

3.2 Inspecting and Adjusting a Function's Configuration

- Click into any function (e.g., `TripWiz-Trips`) from the list.
- On the **Configuration** tab:
 - General configuration** — shows memory allocation and timeout. `TripWiz-Trips` and `TripWiz-TripReminder` are configured with a **60-second timeout** (they perform multiple downstream calls — DynamoDB, Bedrock, SES); all others use the **10-second** default declared in the template's `Globals` section.
 - Environment variables** — shows the runtime configuration injected at deploy time (e.g., `TABLE_NAME` , `BEDROCK_MODEL_ID` , `WS_ENDPOINT` , `DOCUMENTS_BUCKET`). **Do not edit these manually** — they are managed by the SAM template and will be overwritten on the next deployment. To change a value permanently, edit `template.yaml` and redeploy.

Key	
ADMIN_EMAILS	
BEDROCK_MODEL_ID	
DOCUMENTS_BUCKET	
FRONTEND_URL	
MAPPING_SECRET_ARN	
PLACE_INDEX_NAME	
ROUTE_CALCULATOR_NAME	
SOURCE_EMAIL	
TABLE_NAME	
USER_POOL_ID	
VALIDATE_FUNCTION_NAME	
WS_ENDPOINT	

- On the **Permissions** tab, review the function's **execution role** — this defines exactly which AWS resources the function can access (DynamoDB, S3, Bedrock, Secrets Manager, Cognito, Amazon Location Service, etc., as scoped in the template's `Policies` blocks). Never broaden these permissions outside of `template.yaml` — doing so creates configuration drift that will be silently reverted (or cause deployment conflicts) on the next deploy.

3.3 The REST API (API Gateway)

- Open the **API Gateway console** and locate `TripWizApi`.
- This REST API exposes **29 routes** (e.g., `GET /trips`, `POST /trips/{tripId}/optimize`, `GET /admin/users`) — all backed by the single `TripWiz-Trips` Lambda function via proxy integration.
- Click **Stages** → **prod** to view the live deployment stage and its **Invoke URL** (this must match the `TripWizRestApiUrl` value shown in the CloudFormation Outputs).

TripWizApi REST routes (`prod` stage — `https://ln2dwp4q3l.execute-api.us-east-1.amazonaws.com/prod`):

```
/admin
/metrics      GET, OPTIONS
```

/trending	OPTIONS, PUT
/trips	GET, OPTIONS
/{tripId}	DELETE, OPTIONS
/hide	OPTIONS, POST
/users	GET, OPTIONS
/{userId}	DELETE, OPTIONS
/demote	OPTIONS, POST
/promote	OPTIONS, POST
/suspend	OPTIONS, POST
/unsuspend	OPTIONS, POST
/places	
/search	GET, OPTIONS
/routes	
/calculate	OPTIONS, POST
/trending	GET, OPTIONS
/trips	GET, OPTIONS, POST
/{tripId}	DELETE, GET, OPTIONS, PUT
/alerts	GET, OPTIONS
/attachments	GET, OPTIONS
/{fileId}	DELETE, OPTIONS
/complete	OPTIONS, POST
/download-url	GET, OPTIONS
/upload-url	OPTIONS, POST
/collaborators	GET, OPTIONS
/fallbacks	GET, OPTIONS
/invite	OPTIONS, POST
/{userId}	DELETE, OPTIONS
/optimize	OPTIONS, POST
/validate	OPTIONS, POST
/weather	GET, OPTIONS
/user	
/prefs	GET, OPTIONS, PUT

- Under **Authorizers**, confirm `TripWizCognitoAuthorizer` is attached as the **default authorizer** — this enforces that every route (except CORS preflight `OPTIONS` requests) requires a valid Cognito JWT.
- Under **Gateway Responses**, note the custom `DEFAULT_4XX` / `DEFAULT_5XX` responses that inject CORS headers — these exist so that browser clients receive proper CORS headers even on authorization failures and server errors. **Do not remove these.**

3.4 The WebSocket API

- In the API Gateway console, locate `TripWizWebSocketApi` (Protocol type: **WebSocket**).
- Review its **Routes**: `$connect`, `$disconnect`, `join`, `leave`, `edit`, `cursor`, and `ping` — each integrated with `TripWiz-WebSocketHandler` via Lambda proxy.

Routes

Create route

Route selection expression [Info](#)

`$request.body.action`

`$connect`

`$disconnect`

`cursor`

`edit`

`join`

`leave`

`ping`

3. Note that `$connect` is the only route protected by a **custom REQUEST authorizer** (`TripWizWebSocketAuthorizer` , backed by `TripWiz-WebSocketAuthorizer`), which validates the Cognito access token passed as a query-string parameter (`?token=...`) during the WebSocket handshake.
4. Confirm the **Stage** is named `prod` with **Auto-deploy** enabled — this means any route changes pushed via `sam deploy` go live immediately without a manual stage-deployment step.
5. The **WebSocket endpoint URL** (`wss://...execute-api...amazonaws.com/prod`) is shown in the CloudFormation Outputs as `WebSocketEndpoint` — this is the value the frontend uses to establish live-collaboration connections.

4. Managing Data & Storage

4.1 The DynamoDB Table

1. Open the **DynamoDB console** and locate the table **TripWizTable** .
2. Note its configuration:
 - **Billing mode:** `PAY_PER_REQUEST` (on-demand) — there is no provisioned throughput to manage; AWS automatically scales capacity to match load and you are billed per request. This requires **no manual capacity planning**.
 - **Primary key:** Partition key `PK` (string), sort key `SK` (string) — this is a **single-table design** where multiple entity types (`USER#` , `TRIP#` , `CONN#` , `ALERT#` , `ACCESS#` , etc.) coexist in one table, distinguished by key prefixes.
 - **Global Secondary Index GSI1** : keyed on `GSI1PK` + `tripStart` , enabling efficient "trips starting soon" queries (used by the reminder and validation Lambdas).
 - **TTL attribute:** `ttl` — DynamoDB automatically expires and removes items (such as stale WebSocket connection records) once their `ttl` timestamp passes, with **no administrator action required**.

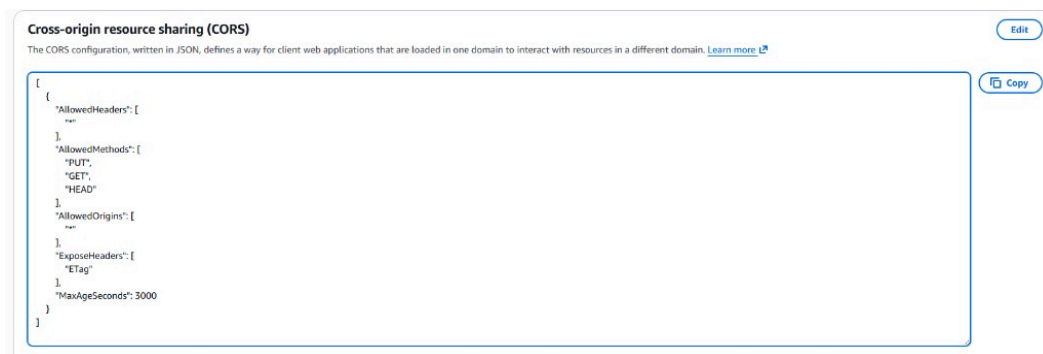
The screenshot shows the 'Global secondary indexes (1/1)' page in the AWS DynamoDB console. It features a table with columns: Name, Status, Partition key, Sort key, Read capacity, Write capacity, Projected attributes, Size, and Item count. The table contains one entry, GSI1, which is Active, has a Partition key of GSI1PK (String) and a Sort key of tripStart (String), with On-demand read and write capacity, All projected attributes, a size of 22.1 kilobytes, and 25 items.

Name	Status	Partition key	Sort key	Read capacity	Write capacity	Projected attributes	Size	Item count
GSI1	Active	GSI1PK (String)	tripStart (String)	On-demand	On-demand	All	22.1 kilobytes	25

3. Under the **Backups** tab, consider enabling **Point-in-Time Recovery (PITR)** if it is not already active — this allows restoring the table to any point in the last 35 days in case of accidental data loss. *(Not enabled by default in the template — this is an operational hardening step the administrator should evaluate.)*
4. Use the **Explore table items** view to inspect data directly when troubleshooting — for example, searching for a specific `USER#{id}` partition to verify a user's trips and preferences.

4.2 The S3 Documents Bucket

1. Open the **S3 console** and locate the bucket named after **TripWizDocumentsBucket** (check the CloudFormation stack's **Resources** tab for its generated name).
2. Review its configuration:
 - **Encryption:** Server-side encryption with AES-256 is enabled by default — all uploaded trip attachments (tickets, vouchers, passes) are encrypted at rest.
 - **Public access:** All four public-access-block settings are enabled (`BlockPublicAcls` , `BlockPublicPolicy` , `IgnorePublicAcls` , `RestrictPublicBuckets`) — **the bucket is never publicly accessible**. Files are only reachable via short-lived presigned URLs generated by `TripWiz-Trips` .
 - **CORS configuration:** Allows `PUT` , `GET` , and `HEAD` from any origin (`*`), exposing the `ETag` header — this is required for the browser-based direct upload/download flow to function.

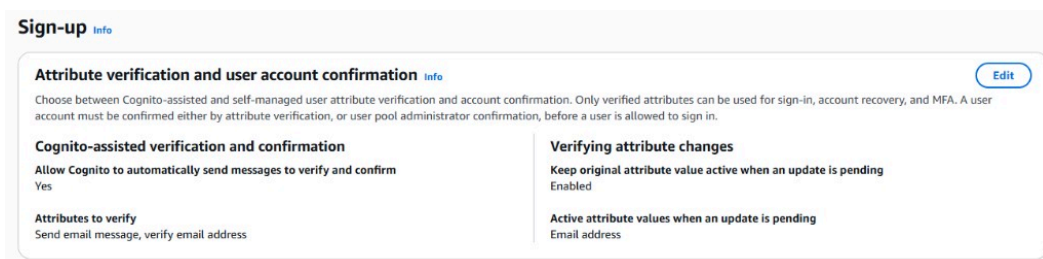


3. **Do not modify the public-access-block or encryption settings** — doing so would create a serious data-exposure risk for users' personal travel documents.
4. Periodically review **storage metrics** (under the **Metrics** tab) to track growth and anticipate costs (cross-reference Section 10 and the Folder 06 cost estimate).

5. Managing Identity & Access (Amazon Cognito)

5.1 The User Pool

1. Open the **Amazon Cognito console** and locate `TripWizUserPool`.
2. Under **Sign-in experience**, confirm `email` is configured as an **auto-verified attribute** — new users must verify their email via a confirmation code before they can sign in (this is the flow documented in Folder 07, Section 1.1).



3. Under **App integration** → **App clients**, locate `TripWizWebClient` :
 - **Client secret**: disabled (`GenerateSecret: false`) — appropriate for a browser-based single-page app, which cannot securely store a secret.
 - **Authentication flows**: `ALLOW_USER_SRP_AUTH` (Secure Remote Password — used for sign-in without ever transmitting the password) and `ALLOW_REFRESH_TOKEN_AUTH` (keeps users signed in across sessions).
 - **PreventUserExistenceErrors** : enabled — sign-in/sign-up error messages never reveal whether a given email is already registered, protecting user privacy.

5.2 The Post-Confirmation Trigger (Default Notification Preferences)

`TripWizUserPool` has a **Lambda trigger** wired to its `PostConfirmation` lifecycle event, pointing at the `TripWiz-PostConfirmation` function (`post_confirmation/index.js`). It fires automatically the moment a new

user verifies their email and their account becomes confirmed — no action is required from the user or an administrator.

1. Under **User pool properties** → **Lambda triggers**, confirm **Post confirmation** is set to `TripWiz-PostConfirmation`.

Post confirmation trigger (live configuration):

Trigger	Lambda function
Post confirmation	TripWiz-PostConfirmation

ARN: `arn:aws:lambda:us-east-1:*****:function:TripWiz-PostConfirmation`

2. **What it does:** the function reads the new user's `sub` (user ID), `email`, `given_name`, and `family_name` from the confirmation event and writes a default **PREFS** record for them to `TripWizTable` (`PK = USER#<sub>`, `SK = PREFS`) with `notifyTrips: true` — i.e. **every newly confirmed user is opted in to trip-reminder and weather-alert emails from the moment they sign up**, without having to first visit the Settings page (Folder 07/08) and save preferences.
3. It uses a conditional write (`attribute_not_exists(PK)`) so it never overwrites an existing **PREFS** record — re-confirmations or retried Cognito events are safe no-ops, and a user who later changes their notification settings keeps their choice.
4. **If a user reports never receiving trip reminders**, check whether their **PREFS** item exists and has `notifyTrips: true` (Section 9 shows how to query `TripWizTable` / read CloudWatch Logs for `TripWiz-PostConfirmation` and `TripWiz-TripReminder`). Accounts created **before** this trigger was deployed will not have a seeded record — they will get one only after their first visit to Settings, or you may backfill one manually via `aws dynamodb put-item`.

5.3 Managing the Admins Group

This is the most important identity-related task for a system administrator: granting or revoking **in-app Admin Portal** access (documented from the user's perspective in Folder 08) at the infrastructure level.

1. In the Cognito console, open `TripWizUserPool` → **Groups** and locate the **Admins** group (precedence: 1).

Admins group members (live configuration):

Username	Status
<code>amitbitton50mail@gmail.com</code>	CONFIRMED
<code>davidkitinberg@gmail.com</code>	CONFIRMED
<code>sagi.hassid@gmail.com</code>	CONFIRMED

2. **To grant admin access to a user:**

- Click **Add user to group**, search for their username/email, and confirm.
- Alternatively, an existing in-app admin can promote the user via the Admin Portal's **Users** page (Folder 08, Section 3) — this calls the `cognito-idp:AdminAddUserToGroup` action under the hood, which the `TripWiz-Trips` function's execution role is explicitly permitted to perform.

3. To **revoke admin access**, select the user within the group and click **Remove user from group** (or use the in-app **Demote** action).
4. **Bootstrap path:** Before any user exists in the `Admins` group, the `AdminBootstrapEmails` stack parameter (set during deployment — see Section 2.2) provides a comma-separated **email whitelist** that is granted admin access as a fallback. Once your team is fully onboarded into the Cognito `Admins` group, you may redeploy with this parameter narrowed down or left as a recovery safety net.

5.4 Resetting a User's Password or Investigating Account Issues

1. From the Cognito console, open **Users**, search for the affected user by email or username, and open their profile.

User information

User ID (Sub)
54989478-a0b1-70bd-1211-eb00606f648

Alias attributes used to sign in
User name

MFA setting
MFA inactive

MFA methods
-

Account status
Enabled

Confirmation status
Confirmed

Created time
May 19, 2025 at 17:36 GMT+3

Last updated time
May 19, 2025 at 17:37 GMT+3

User attributes (2) [Info](#) [Edit](#)

View and edit this user's attributes.

Q Filter by property or value

Attribute name	Value	Type
email	ledude227@gmail.com Verified	Optional
sub	54989478-a0b1-70bd-1211-eb00606f648	Required

2. From here you can view account status (`CONFIRMED` , `UNCONFIRMED` , `FORCE_CHANGE_PASSWORD`), manually confirm a stuck registration, or reset/disable the account directly — useful for support escalations the in-app Admin Portal cannot resolve on its own (e.g., a user who never received their confirmation email).

6. Managing AI & Location Services

6.1 Amazon Bedrock (AI Route Optimization)

1. Open the **Amazon Bedrock console** and confirm the configured model is available in your region. AWS has retired the old standalone **Model access** page; access is now managed through **Model catalog / Inference profiles** (or is enabled by default for serverless models in supported regions).
2. TripWiz invokes **Anthropic Claude Haiku 4.5** via the `BEDROCK_MODEL_ID` environment variable on `TripWiz-Trips` . If optimize requests fail, check CloudWatch logs for `AccessDeniedException` or model-not-found errors (Section 9.3).
3. The exact model identifier the function is configured to invoke is set via the `BEDROCK_MODEL_ID` environment variable (`us.anthropic.claude-haiku-4-5-20251001-v1:0`) — visible on the function's **Configuration** → **Environment variables** tab (Section 3.2). To upgrade to a newer model version, update this value in `template.yaml` and redeploy; do not edit it directly in the console (it will be overwritten on the next deploy).
4. Use **Bedrock's usage metrics** (or CloudWatch, see Section 9) to monitor invocation volume and correlate with the cost projections in Folder 06.

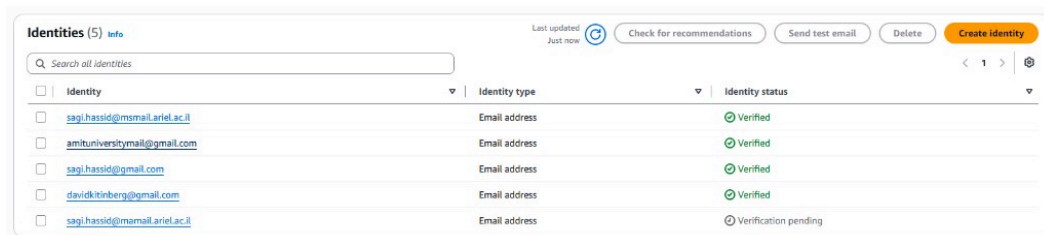
6.2 Amazon Location Service

1. Open the **Amazon Location Service console**.
2. Under **Place indexes**, locate **TripWizPlaceIndex** — confirm its data source is **Esri** and intended use is **SingleUse** (appropriate for live, user-facing geocoding rather than cached/stored results).
3. Under **Routes**, locate **TripWizRouteCalculator** — also backed by **Esri** — used to compute road-following routes between itinerary stops.
4. Both resources are referenced by name via the `PLACE_INDEX_NAME` and `ROUTE_CALCULATOR_NAME` environment variables on `TripWiz-Trips` and `TripWiz-Validate`. Monitor request volume here and compare against the 12-month free-tier allowance discussed in Folder 06 (Section "Amazon Location Service") to anticipate the post-free-tier cost increase.

7. Managing Notifications

7.1 Amazon SES — Sender & Recipient Verification

1. Open the **Amazon SES console** and navigate to **Verified identities**.
2. Confirm that **both** the `SourceEmail` and `AdminEmail` addresses you supplied during deployment (Section 2.2) are listed and show a status of **Verified**.



The screenshot shows the 'Identities (5)' page in the Amazon SES console. It features a search bar, a 'Last updated: Just now' indicator, and buttons for 'Check for recommendations', 'Send test email', 'Delete', and 'Create identity'. Below these is a table with columns for 'Identity', 'Identity type', and 'Identity status'.

Identity	Identity type	Identity status
sagi.hassid@mamail.ariel.ac.il	Email address	Verified
amituniversitymail@gmail.com	Email address	Verified
sagi.hassid@gmail.com	Email address	Verified
davidkittenberg@gmail.com	Email address	Verified
sagi.hassid@mamail.ariel.ac.il	Email address	Verification pending

3. **If your account is in the SES Sandbox** (the default for new AWS accounts), *every* recipient address must also be individually verified before TripWiz can email them — this will block real users from receiving reminder/alert emails in production. Submit a **production access request** from the SES console (**Account dashboard** → **Request production access**) before going live.
4. Monitor the **Reputation dashboard** for bounce and complaint rates — sustained high rates can result in AWS throttling or suspending your sending privileges.

7.2 Amazon SNS — The Alerts Topic

1. Open the **Amazon SNS console** and locate the topic **TripWiz-Alerts**.
2. Confirm that **TripWiz-SnsToSes** is subscribed as a Lambda endpoint — this is the bridge that converts each weather-alert publish (originating from `TripWiz-Validate`) into an email sent to the `AdminEmail` address.

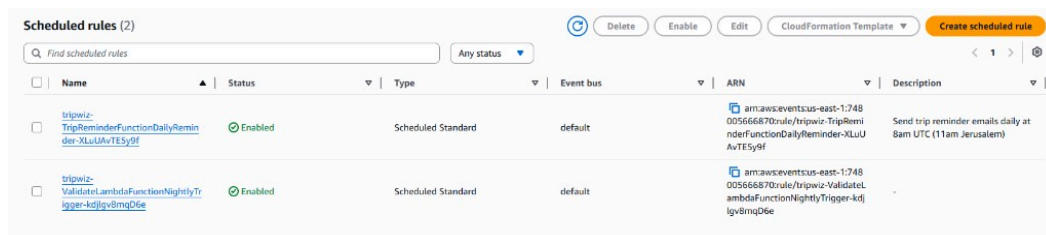
3. Use the **Publish message** test feature here to manually verify the alert→email pipeline end-to-end without waiting for a real weather event.

7.3 Amazon EventBridge — Scheduled Triggers

TripWiz relies on two scheduled (cron/rate-based) triggers:

Schedule Name	Expression	Triggers	Purpose
NightlyTrigger	rate(1 day)	TripWiz-Validate	Re-validates weather for all upcoming trips once daily
DailyReminder	cron(0 8 * * ? *)	TripWiz-TripReminder	Sends reminder emails at 08:00 UTC (11:00 Israel time) daily

1. Open the **Amazon EventBridge console** → **Scheduler** (or **Rules**, depending on how the resource is surfaced) and locate both schedules.



2. To change the reminder timing or frequency, **do not edit the schedule directly in the console** — modify the `Schedule` expression in `template.yaml` (or the `ReminderDays` parameter for the look-ahead window) and redeploy, so the change is versioned and reproducible.
3. Use **Invocation history** (if available in your console view) or CloudWatch Logs (Section 9) to confirm the schedules are firing as expected and the target Lambdas are completing successfully.

8. Managing Secrets (AWS Secrets Manager)

8.1 Why This Matters

The template provisions **two empty secrets** that must be populated before TripWiz is fully operational (see Section 2.4):

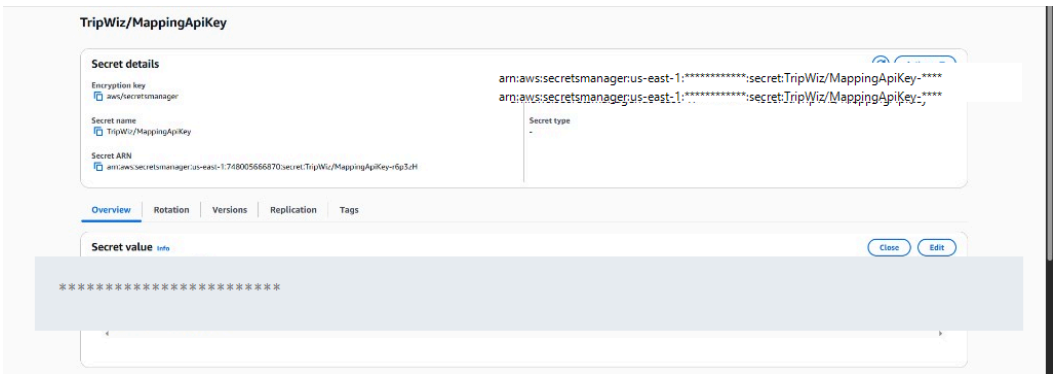
Secret Name	Purpose
TripWiz/OpenWeatherApiKey	API key for the OpenWeatherMap weather-data provider used by TripWiz-Validate

TripWiz/MappingApiKey	API key for any third-party mapping/geocoding provider used alongside Amazon Location Service
-----------------------	---

Both are referenced by `TripWiz-Trips` via the `MAPPING_SECRET_ARN` environment variable, and the function's execution role is granted `secretsmanager:GetSecretValue` **only** for these specific secret ARNs (least-privilege access).

8.2 Populating a Secret's Value

1. Open the **AWS Secrets Manager console** and locate the secret (e.g., `TripWiz/OpenWeatherApiKey`).
2. Click **Retrieve secret value** → **Edit**.



3. Paste the actual API key value obtained from the third-party provider's portal, and click **Save**.
4. Repeat for `TripWiz/MappingApiKey` .

Operational note: Lambda functions cache secret values within their execution environment after the first retrieval (to minimize Secrets Manager API calls and cost — see Folder 06). After **rotating** a secret's value, you may need to wait for existing warm execution environments to recycle (or force this by publishing a new Lambda version) before all functions pick up the new value.

8.3 Rotating Secrets

1. Periodically (e.g., per your organization's security policy, or immediately if a key is suspected to be compromised), repeat the steps in Section 8.2 with a freshly generated API key from the provider.
2. Update the corresponding key in the third-party provider's dashboard to invalidate the old value, ensuring no window where both old and new keys are simultaneously "live" longer than necessary.
3. Confirm functionality by triggering a manual weather validation (via the in-app **Validate Weather** button on a trip, or by manually invoking `TripWiz-Validate` from the Lambda console) and checking CloudWatch Logs for successful API calls.

9. Monitoring, Logging & Troubleshooting

9.1 CloudWatch Log Groups

1. Open the **CloudWatch console** → **Log groups**. Each Lambda function automatically has its own log group, named `/aws/lambda/<FunctionName>` (e.g., `/aws/lambda/TripWiz-Trips`).

Log groups (8)

By default, we only load up to 10,000 log groups.

Filter log groups or try pattern search ☒ Exact match

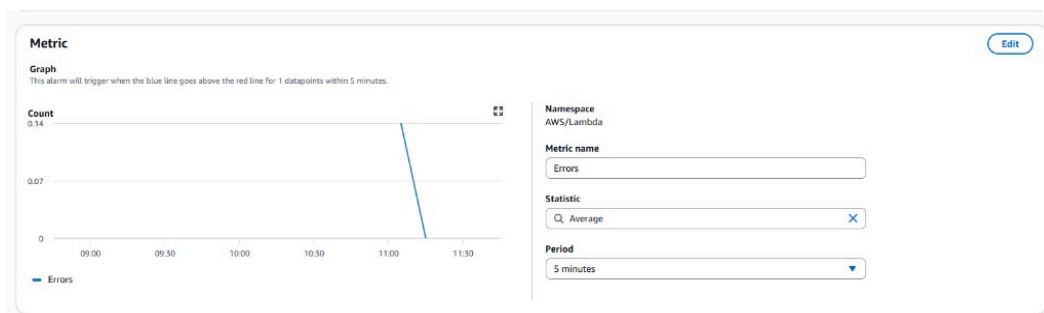
Log group	Log class	Anomaly d...	Deletion pro...	Bearer token...	Data prote...	Sensitive ...	Retention	Metric filters	Co
/aws/lambda/TripWiz-PostConfirmation	Standard	Configure	⊙ Off	⊙ Off	-	-	Never expire	-	⋮
/aws/lambda/TripWiz-PreSignUp	Standard	Configure	⊙ Off	⊙ Off	-	-	Never expire	-	⋮
/aws/lambda/TripWiz-SnsToSes	Standard	Configure	⊙ Off	⊙ Off	-	-	Never expire	-	⋮
/aws/lambda/TripWiz-TripReminder	Standard	Configure	⊙ Off	⊙ Off	-	-	Never expire	-	⋮
/aws/lambda/TripWiz-Trips	Standard	Configure	⊙ Off	⊙ Off	-	-	Never expire	-	⋮
/aws/lambda/TripWiz-Validate	Standard	Configure	⊙ Off	⊙ Off	-	-	Never expire	-	⋮
/aws/lambda/TripWiz-WebSocketAuthorizer	Standard	Configure	⊙ Off	⊙ Off	-	-	Never expire	-	⋮
/aws/lambda/TripWiz-WebSocketHandler	Standard	Configure	⊙ Off	⊙ Off	-	-	Never expire	-	⋮

- Click into a log group and select a recent **log stream** to view individual invocation logs — every function logs key events (e.g., "WebSocket event:", "TripReminder Lambda invoked", "WebSocket authorizer rejected token") that are invaluable for diagnosing issues.
- Use **CloudWatch Logs Insights** to run structured queries across a function's logs — for example, to find all `ConditionalCheckFailedException` (version-conflict) occurrences in `TripWiz-Trips` over the last 24 hours:

```
fields @timestamp, @message
| filter @message like /ConditionalCheckFailedException/
| sort @timestamp desc
```

9.2 Setting Up Alarms

- From any metric view (e.g., a Lambda function's **Monitor** tab, or a log group's **Metric filters**), click **Create alarm**.
- Recommended alarms for TripWiz's production health:
 - Lambda Errors > 0** for `TripWiz-Trips` and `TripWiz-WebSocketHandler` (the two highest-traffic functions) — notify the operator immediately on any failure spike.
 - Lambda Duration approaching Timeout** for `TripWiz-Trips` (60s timeout) — an early warning that downstream calls (Bedrock, DynamoDB, S3) are slowing down.
 - API Gateway 5XX Error Rate** for both `TripWizApi` and `TripWizWebSocketApi`.
 - DynamoDB ThrottledRequests** — though rare with on-demand billing, sustained throttling indicates an extreme traffic spike worth investigating.
- Configure each alarm to publish to an **SNS topic** that emails or pages the on-call administrator.



9.3 Common Issues & Resolutions

Symptom	Likely Cause	Resolution
Users report AI Optimize failing	Bedrock model access not enabled, or <code>BEDROCK_MODEL_ID</code> mismatch	Check Section 6.1; verify model access is Granted in the Bedrock console
Weather validation returns no data	<code>OpenWeatherApiKey</code> secret is empty or invalid	Re-populate the secret per Section 8.2; check <code>TripWiz-Validate</code> logs for HTTP errors from Open-Meteo/OpenWeather
Reminder/alert emails not arriving	SES sandbox mode, unverified sender/recipient, or <code>notifyTrips</code> disabled by the user	Check Section 7.1 (SES verification + production access); confirm the user's notification preference in their Account Settings
Live co-editing not connecting	WebSocket authorizer rejecting tokens, or stale connection records	Check <code>TripWiz-WebSocketAuthorizer</code> logs for "WebSocket authorizer rejected token"; stale <code>CONN#</code> items self-expire via DynamoDB TTL
"Version conflict" errors on trip edits	Two collaborators saved overlapping changes (optimistic concurrency control working as intended)	This is expected behavior — instruct the affected user to refresh and reapply their change. Investigate further only if it occurs persistently for a single trip
410 Gone errors in WebSocket broadcast logs	A connected client disconnected without a clean <code>\$disconnect</code> event	Expected and self-healing — <code>TripWiz-WebSocketHandler</code> automatically deletes stale connection records on 410 responses (see <code>broadcast()</code> in <code>ws_handler/index.js</code>)
New deployment fails with <code>ROLLBACK_COMPLETE</code>	A resource conflict or invalid parameter value	Review the CloudFormation Events tab (Section 2.3) for the specific failing resource and error message; you may need to delete the failed stack before retrying

10. Cost Oversight

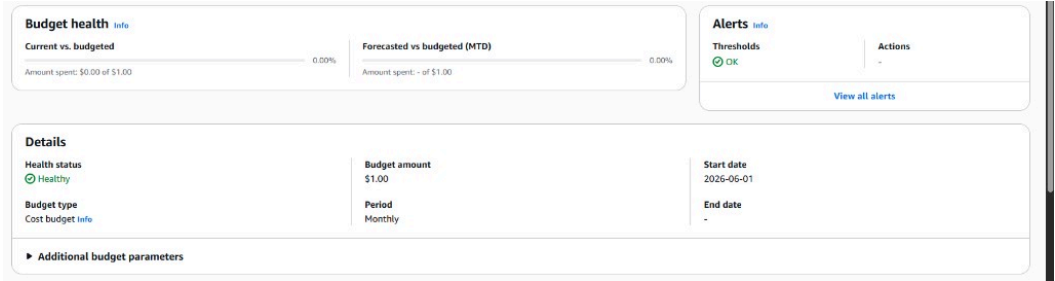
10.1 Where to Monitor Spend

1. Open the **AWS Cost Explorer** console and filter by **service** to see TripWiz's actual monthly spend broken down by Lambda, Bedrock, DynamoDB, Amazon Location Service, and so on. (*Cost Explorer data may take up to 24 hours to appear after first enabling the service — until then, use the projections in [06.md](#).*)
2. Compare actuals against the **detailed projected estimate** in [06.md](#) (Folder 06 — Cost Calculation), which itemizes every service with its expected monthly cost, the official AWS Pricing Calculator export, and a month-by-month year-1 growth projection. Significant deviations from that baseline are an early signal worth investigating (e.g., an unexpected spike in Bedrock invocations could indicate a client-side bug causing repeated optimize calls).

10.2 Setting Up Budget Alerts

1. Open the **AWS Budgets console** and click **Create budget**.

2. Configure a **monthly cost budget** with a threshold informed by the Folder 06 projections (e.g., alert at 80% of the projected ~\$200/month year-1 figure, and again if spend approaches the ~\$313/month steady-state figure).



3. Add an **email/SNS notification** so the administrator is alerted before costs significantly exceed projections — particularly important given that **Amazon Bedrock** and **Amazon Location Service** are the two largest and most usage-sensitive cost drivers identified in Folder 06.